

ISRO plans igloos on the moon

Hoping to eliminate the need for International Space Station, researchers in India are discovering innovative ways to create livable spaces on the moon—the Earth's very own natural satellite. The moon is a wasteland with extreme temperature variations and few resources. The Indian Space Research Organisation (ISRO) researchers plan to build lunar habitats on the moon with an aesthetic very similar to an igloo. These igloos will be constructed using a blend of lunar soil and assorted smart materials that will then

be fed into a 3D printer. This is an innovative approach to building structures in space, as it eliminates the need for materials to be boosted out of Earth's gravitational field with costly rockets.

ISRO hopes these long-term livable habitats will one day house scientists and researchers during their inter-space missions. According to Dr Mylswamy Annadurai, director of the ISRO Satellite Centre in Bangalore, the goal of these structures is ultimately to protect researchers during

their stay and provide a secure place for conducting research.

As an added bonus, by 3D printing the bases, astronauts will no longer have to take on the arduous burden of constructing the buildings themselves while wearing cumbersome space suits. Researchers have successfully demonstrated this project at the space agency's lunar terrain test facility. They have also mastered the creation of a lunar simulant that is 99.6 per cent similar to actual lunar soil.



Lunar settlement (Image credit: Behnaz Parahi/NASA)

Future smart devices to let you feel virtual textures and shapes

Imagine your tablet allowing you to virtually feel the texture and shape of the just-released iPhone from the comforts of your home. Sounds enticing and beats a trip to the nearest Apple store, isn't it? This is the experience that a group of researchers at Northwestern University and France's Aix Marseille University would like you to have in future.

The research team, led by professor Edward Colgate, found that in order for the touchscreens to be able to provide physical cues, it's important to have a reduction in friction between the fingertips and the glass surface of the device. However, the mechanism behind friction reduction has been unknown so far.

To find out how the system works, the scientists performed a study involving seven human participants whose fingers flicked back and forth on a glass plate specially designed



Friction reduction would play a key role in designing smart devices that provide touch feedback (Image credit: <https://pixabay.com/smart-watch-apple-technology-style-821559>)